

# A Logical Phonology of some “minor rules” of Polish

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## 1 Introduction

- Three common stem changes in Polish are illustrated below using partial noun paradigms.

(1) “Yer deletion”:

|    | nom.sg.        | gen.sg.  | gen.pl.   |                |         |
|----|----------------|----------|-----------|----------------|---------|
| a. | kub <b>e</b> ł | kubł-a   | kubł-ów   | ‘bucket’       | (masc.) |
| b. | klusk-a        | klusk-i  | klusek    | ‘noodle’       | (fem.)  |
| c. | zebr-o         | zebr-a   | zeber     | ‘rib’          | (neut.) |
| d. | kufer          | kufr-a   | kufr-ów   | ‘chest, trunk’ | (masc.) |
| e. | szofer         | szofer-a | szofer-ów | ‘driver’       | (masc.) |
| f. | cyfr-a         | cyfr-y   | cyfr      | ‘digit’        | (fem.)  |

(2) “o-raising”:

|    | nom.sg. | gen.sg.          | gen.pl.           |           |         |
|----|---------|------------------|-------------------|-----------|---------|
| a. | stół    | stoł-u           | stoł-ów           | ‘table’   | (masc.) |
| b. | krow-a  | krow-y           | krów              | ‘cow’     | (fem.)  |
| c. | morz-e  | morz-a           | mórz              | ‘sea’     | (neut.) |
| d. | wyrób   | wyro <b>b</b> -u | wyro <b>b</b> -ów | ‘product’ | (masc.) |
| e. | rob-a   | rob-y            | rob               | ‘woman’   | (fem.)  |
| f. | prób-a  | prób-y           | prób              | ‘attempt’ | (fem.)  |

(3) “ $\epsilon$ -backing”:

|    | nom.sg.                            | gen.sg.                               | gen.pl.                                |                   |         |
|----|------------------------------------|---------------------------------------|----------------------------------------|-------------------|---------|
| a. | z <b>ą</b> b                       | z <b>ę</b> b-a                        | z <b>ę</b> b-ów                        | ‘tooth’           | (masc.) |
| b. | r <b>ę</b> k-a                     | r <b>ę</b> k-i                        | r <b>ą</b> k                           | ‘hand, arm’       | (fem.)  |
| c. | świ <b>ę</b> t-o                   | świ <b>ę</b> t-a                      | świ <b>ą</b> t                         | ‘holiday’         | (neut.) |
| d. | ur <b>z</b> ą <b>d</b>             | ur <b>z</b> ę <b>d</b> -u             | ur <b>z</b> ę <b>d</b> -ów             | ‘office’          | (masc.) |
| e. | gr <b>z</b> ę <b>d</b> -a          | gr <b>z</b> ę <b>d</b> -y             | gr <b>z</b> ę <b>d</b>                 | ‘perch, roost’    | (fem.)  |
| f. | pr <b>z</b> yr <b>z</b> ą <b>d</b> | pr <b>z</b> yr <b>z</b> ą <b>d</b> -u | pr <b>z</b> yr <b>z</b> ą <b>d</b> -ów | ‘appliance, tool’ | (masc.) |

- These three patterns—the  $e$ -zero alternation in (1), the  $o$ - $\acute{o}$  (i.e., [ɔ]–[u]) alternation in (2), and the  $\epsilon$ - $q$  ([ $\epsilon\tilde{w}$ ]–[ $\text{ɔ}\tilde{w}$ ]) alternation in (3)—have many things in common:
  - In each, one variant occurs with null desinences (i.e., when the alternating stem vowel is in the final syllable) and another with non-null, vowel-initial desinences.
  - As shown in (1a–c), (2a–c), and (3a–c), the alternations occur across all three genders, even though gender (and animacy) determines where null desinences occur.
  - As shown in (1d–f), (2d–f), and (3d–f), the alternations occur only in a subset of eligible roots even within near-identical phonological contexts; e.g., in (2d–f), not all  $o$ ’s ([ɔ]’s) raise to  $\acute{o}$  ([u]), nor do all [u]’s lower to  $o$ .
- (1–2) are robust patterns found in hundreds of stems, including many likely to be known by children (see Appendix A), whereas (3) is found in no more than a few dozen stems.
- There is much debate about these three alternations, i.e., whether they’re productive, what the locus of exceptionality is, and what rules and representations are implicated.
  - Gussmann (1980, 2007), Rubach (1984, 1986, 2013, 2016), and Spencer (1986) analyze (1) as syncope; Rysling (2016) claims it is epenthesis.
  - Kenstowicz (1994:§2.7.1) and Buckley (2001) provide competing analyses of (2); Sanders (2003, 2006) claims it involves stem suppletion.
  - Westfal (1955) and Bethin (1978:§2) are uncertain whether (3) is better analyzed as backing or fronting.
  - Bethin (op. cit.) refers to all three alternations as “minor rules”.
- I analyze these alternations in Logical Phonology (LP; e.g., Bale and Reiss 2018, Reiss 2021, Gorman and Reiss 2024, Dabbous et al. 2025) illustrating the following themes:
  - Feature-filling processes are implemented using *unification rules*.
  - Segmental deletion can be expressed using *segment rules*.
  - Apparent lexical conditioning of phonological targets is modeled with under- or pre-specification (e.g., Inkelas and Cho 1993, Inkelas 1995) rather than:
    - \* morphophonological rules (e.g., Embick 2012),

- \* morpheme-specific constraints (e.g., Pater 2006, Gouskova 2012, Rysling 2016),
  - \* lexical diacritics (e.g., Lightner 1965, Chomsky and Halle 1968, Kisseberth 1970, Lakoff 1970, Zonneveld 1978), or
  - \* stem suppletion (Sanders 2003, 2006).
- With respect to Polish specifically, LP principles lead us to the following conclusions:
    - (1) is syncope rather than epenthesis,
    - (2) is neither raising nor lowering per se, and
    - (3) is neither backing nor fronting per se.
  - They also motivate a reconsideration of productivity and exceptionality in this domain.
  - Here I focus on nominals—where I am prepared to establish the productivity of these alternations—with occasional supporting evidence from derivatives or other categories.

## 2 Yer deletion

- Let us call any vowel which alternates with zero a *fleeting vowel*.
- Polish fleeting vowels have a plausible historical origin in the Common Slavic sound change called Havlík’s Law, but massive restructuring has occurred in Polish.
- Fleeting *e*’s are most robustly attested in declension (e.g., *kubel*–*kubła* ‘bucket’), but:
  - there are other fleeting vowels, and
  - other categories have fleeting vowels too.
- For example, there are three masculine nouns in which *o* alternates with zero.

(4) *o*–zero alternations (Rubach 2013:1140):

| nom.sg. | gen.sg. | gen.pl. |             |
|---------|---------|---------|-------------|
| kocioł  | kotł-a  | kotł-ów | ‘cauldron’  |
| osioł   | osł-a   | osł-ów  | ‘donkey’    |
| koziół  | kozl-a  | kozl-ów | ‘billygoat’ |

- In the verb ‘swear’, the derived imperfective (DI) inf. *zaklinać* contains a fleeting *i* absent in 3sg. masc. fut. *zaklnie*.
- There are numerous examples which argue against a simple rule of epenthesis or syncope. For instance, (1d–f) shows three possibilities in the context /f\_\_r/:
  - fleeting *e* in *kufer*–*kufra* is consistent with either epenthesis or syncope, but
  - stable *e* in *szofer*–*szofera* seems to contradict syncope, and
  - the absence of *e* in *cyfra*–*cyfry* seems to contradict epenthesis.
- Many, many other near-minimal examples of this form are found in the literature.

## 2.1 Yer epenthesis?

- Rysling (2016), adapting various earlier proposals, claims that *e*-zero alternations reflect epenthesis triggered by morpheme-specific constraints.<sup>1</sup>
- While LP does not necessarily reject morphophonology, it makes a substantive claim that learners prefer narrow phonological analyses *ceteris paribus*.
- With this in mind, consider how Rysling’s analysis might be translated into LP.
- Suppose that the trigger for epenthesis is a +Y feature specification on the segment immediately to the right of the insertion point.<sup>2</sup>
- To express transductions over X-slots in LP, Dabbous et al. (2025) propose a new operator,  $\mapsto$  (`\mapsto` in `LATEX`) and use  $\epsilon$  to symbolize nullity.
- So far we have seen fleeting *e* surfacing in a final syllable, suggesting the following rule.

(5) *e*-epenthesis:

$$\epsilon \mapsto \left\{ \begin{array}{c} \text{X} \\ | \\ \begin{array}{l} -\text{HIGH} \\ -\text{LOW} \\ -\text{BACK} \\ \dots \end{array} \end{array} \right\} / \text{ — } [+Y] \%$$

- Whatever feature Y is, +Y is presumably deleted across the board by a later rule.

(6)  $[ ] \setminus \{+Y\}$

- I suspect many find this analysis intuitively distasteful. Here’s one of its many empirical problems: fleeting *e* unexpectedly surfaces in the diminutive *kuferek* (\**kufrek*).
- There are ways one might rescue the epenthesis analysis at this point, but Gussmann (1980:§2) and Rubach (2016) give many other arguments against it, so I put it aside.

<sup>1</sup>Bethin (op. cit.:§1.20–23), citing Brooks (1975), suggests some fleeting vowels are epenthesized whereas others are present underlyingly and undergo syncope; I see no advantage for this split analysis under my more modest assumptions.

<sup>2</sup>An alternative analysis in which the trigger is to the immediate left of the insertion point seems unworkable. There are suffixes—including the diminutive *-k/-ek* and the adjective-forming *-n/-en*—with fleeting vowels despite the absence of a tautomorphic segment to the left of the insertion point.

## 2.2 Yer syncope!

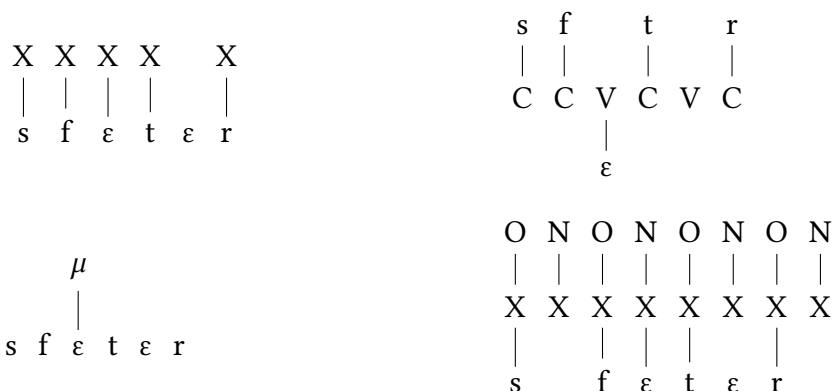
- Rubach (2013) argues against an account in which fleeting vowels are not representationally distinct from stable mid vowels but undergo morphologically-conditioned syncope (à la Gouskova 2012 on Russian):
  - many stems contain multiple mid vowels (e.g., *sweter–swetra* ‘sweater’),
  - and given the language’s relatively permissive phonotactics,
  - it is difficult to predict which will or will not undergo syncope.
- This leads some authors to claim that Polish fleeting vowels are underlyingly present but are representationally distinct from non-fleeting vowels.
- For example, Gussmann (1980) and Rubach (1984)—adapting Lightner’s (1965)’s analysis of Russian—assume that fleeting *e* is underlyingly +HIGH and –TENSE:
  - +HIGH contrasts fleeting *e* with non-fleeting [ɛ].
  - –TENSE contrasts fleeting *e* with *i* [i] and *y* [ɨ].
- I will initially assume the following underlying vowel inventory/specification.

(7) Vowel specification (to be revised):

|       | i | ɨ | u | ĩ | ĭ | ɛ | ɔ | a |
|-------|---|---|---|---|---|---|---|---|
| HIGH  | + | + | + | + | + | – | – | – |
| LOW   | – | – | – | – | – | – | – | – |
| TENSE | + | + | + | – | – | – | – | – |
| BACK  | – | + | + | – | + | – | + | + |
| ROUND | – | – | + | – | – | – | + | – |

- High lax /ĩ, ĭ/—called the *yers* in traditional grammar—never surface as such.
- The /ĩ ~ ĭ/ contrast plays no role here but is important elsewhere in the grammar.
- In contrast, later work tends to treat the fleeting vowels as prosodically deficient:
  - Rubach (1986) proposes they lack an underlying X slot.
  - Spencer (1986) proposes they are bare V slots.
  - Rubach (2016) proposes they lack an underlying mora.
  - Gussmann (2007) proposes they are not linked to a nuclear X slot.

(8) Prosodically deficient representations of *sweter* ‘sweater’:



- Recall that LP segmental rules cannot refer to the absence of a feature specification.
- In a similar spirit, I propose LP autosegmental rules cannot refer to the absence of an association line between some node  $n$  on tier  $T$  and any other node on tier  $T'$  ( $\neq T$ ).
- Conditions like “/ε/ not associated with an X slot” are just one of a possibly infinite set of logical predicates **not** found in LP.
- This proposal, however, makes it impossible to target just the fleeting vowel (i.e., the second /ε/) in representations like (8):
  - [⊕] will match any /ε/, even those associated with an X slot or mora.
  - [V] will match any vowel, even those associated with a segmental node.
- Shortly, I will argue that fleeting (but not stable) vowels are triggers in addition to targets, and the same issues arise in that case.
- Given this, I favor the earlier proposal that the fleeting vowels are the featurally (rather than prosodically) distinct /ĩ, ĩ/, and that:
  - they are both the target and trigger of rules allowing the fleeting  $e$  to surface,
  - they trigger  $o$ -raising, and
  - they trigger  $e$ -backing.
- Following Gussmann (op. cit.), henceforth G, I propose that the null desinences (e.g., the masc. nom.sg. and fem./neut. gen.pl.) are underlyingly /-ĩ/.
- Then the basic generalization, going back to G, is as follows:
  - yers surface as  $e$  if there is a yer in the following syllable,
  - and delete otherwise.

- The following two rules map the yers to /ε/:<sup>3</sup>

$$(9) \left[ \begin{array}{c} +\text{HIGH} \\ -\text{TENSE} \end{array} \right] \setminus \left\{ \begin{array}{c} +\text{HIGH} \\ +\text{BACK} \end{array} \right\} / \text{--- } C_0 \left[ \begin{array}{c} +\text{HIGH} \\ -\text{TENSE} \end{array} \right]$$

$$(10) [ ] \sqcup \left\{ \begin{array}{c} -\text{HIGH} \\ -\text{BACK} \end{array} \right\}$$

- Then, the following context-free rule deletes the remaining yers.

(11) Yer deletion:

$$\begin{array}{c} X \\ | \\ \left[ \begin{array}{c} +\text{HIGH} \\ -\text{TENSE} \end{array} \right] \mapsto \epsilon \end{array}$$

(12) Critical ordering: (9) << (10–11)

- Under the assumption that the diminutive is an /-ĭk/ suffix followed by the ordinary case/number suffixes, the above rules correctly predict:
  - fleeting stem *e*'s appearance throughout the diminutive paradigm, and
  - the distribution of the two allomorphs of the diminutive.

(13) Full paradigms of *kufer* and *kuferek*:

|      | sg.           | pl.      | sg.                  | pl.                  |
|------|---------------|----------|----------------------|----------------------|
| nom. | ku <b>fer</b> | kufr-y   | ku <b>fer-ek</b>     | ku <b>fer</b> -k-i   |
| gen. | kufr-a        | kufr-ów  | ku <b>fer</b> -k-a   | ku <b>fer</b> -k-ów  |
| dat. | kufr-owi      | kufr-om  | ku <b>fer</b> -k-owi | ku <b>fer</b> -k-om  |
| acc. | ku <b>fer</b> | kufr-y   | ku <b>fer-ek</b>     | ku <b>fer</b> -k-i   |
| ins. | kufr-em       | kufr-ami | ku <b>fer</b> -k-iem | ku <b>fer</b> -k-ami |
| loc. | kufr-ze       | kufr-ach | ku <b>fer</b> -k-u   | ku <b>fer</b> -k-ach |
| voc. | kufr-ze       | kufr-y   | ku <b>fer</b> -k-u   | ku <b>fer</b> -k-i   |

(14) Sample derivations for *kufer* and *kuferek*:

|              | nom.sg.   | gen.sg.   | nom.sg. dim. | gen.sg. dim. |
|--------------|-----------|-----------|--------------|--------------|
| UR           | /kufĭr-ĭ/ | /kufĭr-a/ | /kufĭr-ĭk-ĭ/ | /kufĭr-ĭk-a/ |
| Rule (9–10): | kuferĭ    |           | kuferekĭ     | kuferĭka     |
| Rule (11):   | kufer     | kufra     | kuferek      | kuferka      |
| ...          |           |           |              |              |
| SR           | [kufer]   | [kufra]   | [kuferek]    | [kuferka]    |

<sup>3</sup>For concreteness, I assume (9) applies simultaneously (à la *SPE*) or, equivalently here, left-to-right.

### 3 o-raising

- The *o-ó* alternation (e.g., *stół-stołu* ‘table’) is also common in nouns.

|         | nom.sg. | gen.sg.          | gen.pl.           |           |
|---------|---------|------------------|-------------------|-----------|
| a.      | stół    | stoł-u           | stoł-ów           | ‘table’   |
| b.      | krow-a  | krow-y           | krów              | ‘cow’     |
| (15) c. | morz-e  | morz-a           | mórz              | ‘sea’     |
| d.      | wyrób   | wyro <b>b</b> -u | wyro <b>b</b> -ów | ‘product’ |
| e.      | rob-a   | rob-y            | rob               | ‘woman’   |
| f.      | prób-a  | prób-y           | prób              | ‘attempt’ |

- For stems that exhibit *o*-raising, raised *ó* occurs throughout the diminutive paradigm.

(16) “*o*-raising” in diminutives:

| nom.sg. | gen.sg.        | gen.pl.         | dim.              |                  |
|---------|----------------|-----------------|-------------------|------------------|
| wóz     | w <b>o</b> z-u | w <b>o</b> z-ów | w <b>ó</b> z-ek   | ‘wagon’ (masc.)  |
| wdow-a  | wdow-y         | wd <b>ó</b> w   | wd <b>ó</b> w-k-a | ‘widow’ (fem.)   |
| drog-a  | drog-i         | dr <b>ó</b> g   | dr <b>ó</b> ż-k-a | ‘drug’ (fem.)    |
| koł-o   | koł-a          | k <b>ó</b> ł    | k <b>ó</b> ł-k-o  | ‘circle’ (neut.) |

- There are a few doublets: e.g., the gen.pl. of *zmora* ‘nightmare’ is either *zmór* or *zmor* (Gussmann 2007:265).
- It is also found in several other categories, including pronouns and adjectives.

(17) *o*-raising in nom.sg. pronouns:

| masc.vir.     | masc.nvir.      |                     |
|---------------|-----------------|---------------------|
| m <b>ó</b> j  | m <b>o</b> j-e  | 1sg. poss. (‘my’)   |
| tw <b>ó</b> j | tw <b>o</b> j-e | 2sg. poss. (‘your’) |
| sw <b>ó</b> j | sw <b>o</b> j-e | refl. poss.         |

(18) *o*-raising in short-form adjectives:

| long form | short form     |           |
|-----------|----------------|-----------|
| gotow-y   | got <b>ó</b> w | ‘ready’   |
| wesoł-y   | wes <b>ó</b> ł | ‘merry’   |
| zdrow-y   | zdr <b>ó</b> w | ‘healthy’ |

- The imperative of some verbs shows *o*-raising as in (19ab) though others fail to undergo it as in (19cd) or exhibit doublets as in (19ef).<sup>4</sup>

<sup>4</sup>I have been unable to find any specific information about the nature of this variation, i.e., whether it is intra-speaker, inter-speaker, or both. The same is true of other doublets reported elsewhere in this talk.



(19) *o*-raising in verbs (Buckley 2001 passim):

|    | 1sg. impf. | imper.      |           |
|----|------------|-------------|-----------|
| a. | godz-ę     | gódź        | ‘agree’   |
| b. | robi-ę     | rób         | ‘do’      |
| c. | chodz-ę    | chodź       | ‘walk’    |
| d. | skrobi-ę   | skrob       | ‘scratch’ |
| e. | ogol-ę     | ogol ~ ogól | ‘shave’   |
| f. | woż-ę      | woż ~ wóz   | ‘carry’   |

- When the alternating vowel is followed by a stem-final voiced consonant:
  - *ó* surfaces when this consonant is word-final (and devoiced), and
  - *o* appears otherwise.
- This pattern has a plausible historical origin—the reanalysis of compensatory lengthening after the early 16th century loss of the length contrast (see Bethin 1978:§2.11 and Buckley 2001)—but that’s irrelevant to the synchronic analysis.
- Kenstowicz (1994), henceforth K, claims the alternating vowel is /ɔ/ and that raising to [u] is triggered by a following word-final +VOICE consonant.<sup>5</sup>

(20) Raising (after K):

$$\begin{bmatrix} -\text{HIGH} \\ -\text{LOW} \\ +\text{BACK} \end{bmatrix} \rightarrow \{+\text{HIGH}\} / \text{---} \begin{bmatrix} +\text{CONSONANTAL} \\ +\text{VOICE} \\ -\text{NASAL} \end{bmatrix} \%$$

- *Roba-robby-rob* ‘woman’ in (2e) and the following are thus *negative exceptions* to (20).

|      | nom.sg.  | gen.sg.  | gen.pl.   |             |         |
|------|----------|----------|-----------|-------------|---------|
|      | drodz    | drodz-a  | drodz-ów  | ‘thrush’    | (masc.) |
| (21) | batog    | batog-a  | batog-ów  | ‘whip’      | (masc.) |
|      | żywiół   | żywiół-u | żywiół-ów | ‘element’   | (masc.) |
|      | sikor-a  | sikor-y  | sikor     | ‘chickadee’ | (fem.)  |
|      | jezior-o | jezior-o | jezior    | ‘lake’      | (neut.) |

- Under a traditional approach to exceptionality (e.g., Chomsky and Halle 1968:§8.7, Zonneveld 1978:§3), there is, for each rule *R*, a rule feature  $\pm R$  subject to the following.

(22) *SPE* rule feature conventions:

<sup>5</sup>While K does not provide relevant examples, raising triggers in his analysis presumably include liquids (e.g., *pozór-pozoru-pozorów* ‘illusion’) but not nasals, and (20) has been modified to reflect this.

- a. If a morpheme is not specified  $-R$ , introduce  $+R$  via redundancy rule.
  - b. If a morpheme is specified  $\alpha R$ , propagate  $\alpha R$  to each of the segments in its UR.
  - c. Rule  $R$  does not apply to  $-R$  segments.
- These conventions suffice to generate negative exceptions: /rɔb-/ and morphemes like it can be simply entered into the lexicon as  $-RAISING$ .
  - However, K's account also has *positive exceptions*, because the raised  $\acute{o}$  is also found before underlyingly voiceless consonants.

|      | nom.sg.            | gen.sg.            | gen.pl.            |          |         |
|------|--------------------|--------------------|--------------------|----------|---------|
|      | powró <b>t</b>     | powro <b>t</b> -u  | powro <b>t</b> -ów | 'return' | (masc.) |
| (23) | cn <b>o</b> t-a    | cn <b>o</b> t-y    | cn <b>o</b> t      | 'virtue' | (fem.)  |
|      | rob <b>o</b> t-a   | rob <b>o</b> t-y   | rob <b>o</b> t     | 'work'   | (fem.)  |
|      | sio <b>o</b> str-a | sio <b>o</b> str-y | si <b>o</b> str    | 'sister' | (fem.)  |
|      | stop <b>o</b> -a   | stop <b>o</b> -y   | st <b>o</b> p      | 'foot'   | (fem.)  |

- Following (22), marking /pɔvrɔt-/ (etc.) as underlyingly  $+RAISING$  will not suffice to trigger the rule, since the root still will not match the phonological context for (20).
- K's account also has no explanation for the presence of raised  $\acute{o}$  in diminutives, which constitute a large class of positive exceptions to (20).
- There are various ways one could extend the *SPE* conventions to handle positive exceptions but I put this aside for now.
- These exceptions lead Buckley (2001), henceforth B, to reject K's analysis. My analysis is a loose LP translation of B's approach.
- B proposes that alternating vowels, denoted by /O/, are underspecified for HIGH; I also leave it underspecified for TENSE.

(24) Vowel specification (partial, revised):

|       | ɔ | O | u |
|-------|---|---|---|
| HIGH  | — |   | + |
| LOW   | — | — | — |
| TENSE | — |   | + |
| BACK  | + | + | + |
| ROUND | + | + | + |

- A unification rule targets vowels followed by a yer in the next syllable, regardless of the voicing of the following consonant, non-vacuously mapping /O/  $\rightsquigarrow$  /u/ in this context.

$$(25) \begin{bmatrix} -\text{Low} \\ +\text{BACK} \end{bmatrix} \sqcup \begin{Bmatrix} +\text{HIGH} \\ +\text{TENSE} \end{Bmatrix} / \text{--- } C_0 \begin{bmatrix} +\text{HIGH} \\ -\text{TENSE} \end{bmatrix}$$

- A later context-free redundancy rule non-vacuously maps /O/  $\rightsquigarrow$  /ɔ/ elsewhere.

$$(26) \begin{bmatrix} -\text{Low} \\ +\text{BACK} \end{bmatrix} \sqcup \begin{Bmatrix} -\text{HIGH} \\ -\text{TENSE} \end{Bmatrix}$$

(27) Critical ordering: (25)  $\ll$  (9, 26)

- (25–26) vacuously apply to the fully-specified, non-alternating /ɔ, u/.

(28) Sample derivations:

|            | ‘cow’ (nom.sg.) | ‘cow’ (gen.pl.) | ‘woman’ (gen.pl.) | ‘attempt’ (gen.pl.) |
|------------|-----------------|-----------------|-------------------|---------------------|
| UR         | /krOv-a/        | /krOv-ĩ/        | /rɔb-ĩ/           | /prub-ĩ/            |
| Rule (25): |                 | kruv            |                   |                     |
| Rule (26): | krɔva           |                 |                   |                     |
| ...        |                 |                 |                   |                     |
| SR         | [krɔva]         | [kruf]          | [rɔp]             | [prup]              |

- If, as previously assumed, the diminutive suffix is /-ĩk/ followed by ordinary case/number desinences, these rules also correctly predict the raised ó throughout the diminutive.<sup>6</sup>

## 4 ɛ-backing

- The rarer ɛ–q alternations in nouns (e.g., *zɛb–zɛba* ‘tooth’) can be given a similar analysis to *o–ó* alternations.

|      | nom.sg.  | gen.sg.    | gen.pl.     |                   |
|------|----------|------------|-------------|-------------------|
|      | zɛb      | zɛb-a      | zɛb-ów      | ‘tooth’           |
|      | rɛk-a    | rɛk-i      | rɛk         | ‘hand, arm’       |
| (29) | święt-o  | święt-a    | święt       | ‘holiday’         |
|      | urząd    | urząd-u    | urząd-ów    | ‘office’          |
|      | grzęd-a  | grzęd-y    | grzęd       | ‘perch, roost’    |
|      | przyrząd | przyrząd-u | przyrząd-ów | ‘appliance, tool’ |

<sup>6</sup>There are some diminutives of *o*-raising nouns, mostly masculines, which fail to show the raised ó, and a few doublets (e.g., *ogrodek* vs. *ogródek* < *ogród–ogrodu* ‘garden’). According to Gussmann, “if there are two forms (with and without Raising), it is usually the one without Raising that sounds obsolete” (G:119). I tentatively propose that non-raised diminutives are built from a separate root; i.e., that *ogrodek* is not built from the /ogrOd-/ root of *ogród*. In some the “doublets” are semantically distinct; e.g., *żłobek* ‘creche’ vs. *żłódek* ‘crib’ < *żłób–żłobu* ‘manger’.

- The so-called nasal vowels  $\eta$  and  $\epsilon$  appear to be underlyingly oral mid vowels of some sort followed by a placeless nasal I'll write as /N/.
- Evidence for this biphonemic analysis comes from verbs (Gussmann 1980:§3, Rubach 1984:§5.6) where a fleeting vowel in (30ab) surfaces as a nasal vowel as in (30cd).

(30) Stem allomorphy in /za-klĩn-/ 'swear':

- a. zaklinać (DI) impf. inf.
- b. zaklnie 3sg. masc. fut.
- c. zakleła perf. inf.
- d. zakląć 3sg. masc. perf.

- The exact surface form of the nasal vowels is disputed, but roughly they are:
  - [ɛ̃, ɔ̃] word-finally (e.g., *lwię* [lvjɛ̃] 'lion cub') and before fricatives (e.g., *wąs* [vɔ̃s] 'mustache'), and
  - oral mid vowels followed by a homorganic nasal stops elsewhere (e.g., *kępa* [kɛ̃pa] 'cluster', *kolęda* [kɔ̃lɛ̃nda] 'carol', *mąka* [mɔ̃ŋka] 'flour').
- I note there are a number of nouns exhibiting doublets with respect to this alternation.

(31)  $\epsilon$ -backing doublets (Rubach 1986:145):

| nom.sg.         | gen.sg.   | gen.pl.       |          |         |
|-----------------|-----------|---------------|----------|---------|
| swęd ~ swąd     | swęd-u    | swąd-ów       | 'stench' | (masc.) |
| wstęg-a         | wstęg-i   | wstęg ~ wstąg | 'ribbon' | (fem.)  |
| żołędź ~ żołądź | żołędzi-a | żołędz-i      | 'acorn'  | (masc.) |

- A rule which simply backs any  $\epsilon$  in a final syllable would have many negative exceptions; e.g., *grzęda*–*grzęd-y*–*grzęd* in (3e) and the following.

|      | nom.sg. | gen.sg. | gen.pl. |             |         |
|------|---------|---------|---------|-------------|---------|
| (32) | oręż    | oręż-a  | oręż-ów | 'weapon'    | (masc.) |
|      | klęp-a  | klęp-y  | klęp    | 'moose cow' | (fem.)  |
|      | mięs-o  | mięs-a  | mięs    | 'meat'      | (neut.) |

- Let /E/ denote a mid vowel underspecified for BACK, the target for this alternation.

(33) Vowel specification (partial, revised):

|       | ɛN | ɔN | EN |
|-------|----|----|----|
| HIGH  | –  | –  | –  |
| LOW   | –  | –  | +  |
| TENSE | –  | –  | –  |
| BACK  | –  | +  |    |
| ROUND | –  | –  | –  |

- In alternating stems:
  - the +BACK alternant  $q$  is found in final syllables when followed by a yer desinence and throughout the paradigms of diminutives—cf. (13) and 16)—and
  - the –BACK alternant  $ɛ$  occurs elsewhere.

|      | nom.sg.          | gen.sg.          | gen.pl.         | dim.              |             |
|------|------------------|------------------|-----------------|-------------------|-------------|
| (34) | z <b>ą</b> b     | z <b>ɛ</b> b-a   | z <b>ɛ</b> b-ów | z <b>ą</b> b-ek   | ‘tooth’     |
|      | r <b>ę</b> k-a   | r <b>ę</b> k-i   | r <b>ą</b> k    | r <b>ą</b> cz-k-a | ‘hand, arm’ |
|      | świ <b>ę</b> t-o | świ <b>ę</b> t-a | świ <b>ą</b> t  | świ <b>ą</b> t-ek | ‘holiday’   |

- My analysis is a loose LP translation of earlier proposals by Gussmann (1980:§3) and Rubach (1984:§5.6.4).
- A unification rule targets pre-nasal mid vowels followed by a yer in the next syllable, non-vacuously mapping /EN/  $\rightsquigarrow$  /ɔN/ in this context.

$$(35) \left[ \begin{array}{c} -\text{HIGH} \\ -\text{LOW} \end{array} \right] \sqcup \{+\text{BACK}\} / \text{ — } [+ \text{NASAL}] \text{ C}_0 \left[ \begin{array}{c} +\text{HIGH} \\ -\text{TENSE} \end{array} \right]$$

- A later redundancy rule non-vacuously maps /EN/  $\rightsquigarrow$  /ɛN/ elsewhere.

$$(36) \left[ \begin{array}{c} -\text{HIGH} \\ -\text{LOW} \end{array} \right] \sqcup \{-\text{BACK}\} / \text{ — } [+ \text{NASAL}]$$

(37) Critical ordering: (35)  $\ll$  (9, 36)

- Subsequent post-lexical rules derive the surface forms described above.

(38) Sample derivations for **z**ą**b**:

|            | nom.sg.  | gen.sg.  |
|------------|----------|----------|
| UR         | /zENb-ĩ/ | /zENb-a/ |
| Rule (35): | zɔNbĩ    |          |
| Rule (36): |          | zɛNba    |
| ...        |          |          |
| SR         | zɔmp     | zɛmba    |

## 5 Conclusions

- LP permits an analysis of these patterns within the narrow phonology, without reference to morphosyntactic context or lexical identity and without suppletion.
- I am pushing the narrow phonology as far as possible, under the hypothesis that children acquiring language are *epistemically bounded* (in the sense of Fodor 1980:33f.) to prioritize narrow phonological solutions.
- The analyses given here are essentially LP translations of earlier analyses by Buckley, Gussmann, and Rubach, but these translations are non-trivial.
  - LP requires a traditional, featurally-distinct representation of the fleeting vowels (*pace* Gussmann, Rubach, and Spencer) and in this way is more constrained.
  - LP does not require a feature-filling/feature-changing rule diacritic (*pace* Buckley).
  - LP challenges distinctions like “raising” vs. “lowering” or “fronting” vs. “backing”.
- These analyses lead to a reconsideration of productivity/exceptionality in this domain:
  - Sanders (2003, 2006) takes the productivity of *o*-raising to be a question about the productivity of morphophonological rules (cf. Sneller et al. 2019).
  - But in my analyses, the relevant phonological rules are all exceptionless.
  - Thus, the under- or over-application of these patterns in, e.g., child language errors or *wug*-tasks involves generalizations over URs (cf. Schütze 2005, Scheer 2019).

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## A Type counts

To get a rough estimate of the type frequencies of the three stem changes, I extracted the 1,000 most frequent noun stems from a frequency list for child-directed Polish, compiled by Ewa Haman and colleagues from published (Smoczyńska 1985, Weist et al. 1984, Weist and Witkowska-Stadnik 1986) and unpublished corpora.<sup>7</sup> Full paradigms for these nouns were then obtained using the online edition of the *Grammatical Dictionary of Polish* (Saloni et al. 2015, Woliński and Kieraś 2016).<sup>8</sup> The allomorphs of these forms were then auto-coded using Pynini (Gorman and Sproat 2021) finite-state grammars. For each candidate rule, I report:

- *n*, the number of stems which undergo the alternation,
- *e*, the number of stems which do not undergo the alternation but which otherwise meet the structural description,
- whether the majority of stems participate in the pattern, and
- whether the pattern is *sufficient* in the sense of Yang (2016).

To provide sensible counts, it is necessary to reformulate the rules in a surface-oriented form, as is done below. Results are shown in Table 1.

<sup>7</sup><https://childes.talkbank.org/access/Slavic/Polish/Polish-CDS.html>

<sup>8</sup><http://sgjp.pl/>



|                           | $n$ | $e$ | majority? | sufficient? |
|---------------------------|-----|-----|-----------|-------------|
| (39): yer deletion        | 340 | 51  | ✓         | ✓           |
| (40): $o$ -raising        | 40  | 12  | ✓         | ✓           |
| (41): $\epsilon$ -backing | 4   | 8   | ✗         | ✗           |

Table 1: Counts of the three stem change patterns (at  $r \leq 1,000$ ).

(39) Yer deletion (surface-oriented):

$$\epsilon \rightarrow \emptyset / \text{--- } C_0 \%$$

(40)  $o$ -raising (surface-oriented):

$$\mathfrak{u} \rightarrow \mathfrak{u} / \text{--- } \left[ \begin{array}{c} +\text{CONSONANTAL} \\ +\text{VOICE} \\ -\text{NASAL} \end{array} \right] \%$$

(41)  $\epsilon$ -backing (surface-oriented):

$$\epsilon \rightarrow \mathfrak{u} / \text{--- } [+ \text{NASAL}] C_0 \%$$